

UNIFIED MECHANICS · THE MONOGRAPH

A World of Distinctions

How a sentence, asked to describe itself, becomes a universe.

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On the grades. *Claims are kept in grades throughout, and never quietly mixed. PROVED follows from the stated assumptions. CONDITIONAL follows once a named condition is granted. PROPOSED is a physical reading offered for calculation and falsification. POSTULATE is a starting commitment. The difference between a framework and a mood is that a framework keeps these apart.*

MOVEMENT ONE

The one move

Consider a sentence. Not the idea of a sentence. This one, here, presently doing its job. It begins and it ends, which gives it an edge. It is built from words, which gives it parts. The words arrive in an order, so they resolve one after another. And it means something, though the meaning lives in none of the words alone. The meaning is the whole they settle into. Before a sentence is anything so dignified as literary, it is a system: differences, held together by relations, following a rule.

Ask the sentence to do one small thing. Ask it to describe itself.

That request is the whole of this book. The golden ratio, the three weights, light and matter, the boundary that nothing sees past: all of it unfolds from that single request, each step following from the last, with nothing slipped in by hand along the way.

MOVEMENT TWO

What a thing has to do to be a system

It is worth watching, first, what anything at all must do before the word system honestly applies to it.

It makes a difference. This is not optional. Until there is one difference somewhere, there is nothing for a system to be made of, and the whole enterprise is off to a slow start. Having made the difference, the thing has to hold it, keeping an inside apart from an outside, or the difference simply leaks away and is not heard from again. Held, the difference settles into a single thing. The single thing answers to wherever it happens to find itself. Answering, it changes, though only in the ways its edge permits. One change follows another, and in that following a direction quietly appears. And every change it passes through leaves something behind that can be read later.

distinguish · hold · close · answer · change · order · record

Seven doings, arranged in the only order they can occur in, since nothing is held before it is drawn and nothing recorded before it has happened. Read down the list, and it is not an inventory of parts. It is the shortest story anywhere of how a thing comes to last.

Something is conspicuously missing from that story: things. There are only doings. A part, an edge, a record, each turns out to be a doing that has finished. A noun, put plainly, is a verb that has stopped moving. The point is not decorative. It keeps the account honest. Anywhere this book appears to name a thing, it is naming a completed action, and the action can, in principle, be set going again.

MOVEMENT THREE

The loop that starts everything

A stone manages six of the seven without visible effort, having nowhere in particular to be. The seventh is where the trouble starts, because a record can be read, and the thing reading it can be the very thing that wrote it.

When a system turns around to describe itself, it meets a difficulty that never troubles an outside observer. It has to hold two things at once: what it is presently working out, and what it must keep in reserve in order to carry on. A single channel cannot manage this. With one channel, describing oneself amounts to writing over oneself, which rather defeats the purpose. Two channels are needed. One carries what is resolved now. The other carries what is held back for later.

That is the entire seed. Two channels, and one rule for how they move forward. The next resolved part is the present resolved part together with what was held. The next reserve is simply the present resolved part, promoted. Written as one step:

$$(\text{resolved}, \text{held}) \text{ becomes } (\text{resolved} + \text{held}, \text{resolved})$$

Everything after this is consequence.

MOVEMENT FOUR

The golden proportion

The state is a pair of numbers: how much is resolved, how much is held. The rule carries the pair forward. To see what it forces, it helps to ask which rules are even allowed, and to be as unambitious as possible about it.

Ask only for the plainest update imaginable. It should add and never subtract, since a system describing itself is not in the business of losing its own content. It should carry exactly one clean copy of what is resolved and one clean copy of what is held into the next resolved amount, no more and no less. And the next reserve should be exactly the present resolved amount, with nothing clever done to it. These are modest requests. They are also, as it happens, enough. Precisely one update satisfies all of them:

$$\begin{bmatrix} 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \end{bmatrix}$$

That small square is the hinge the whole book turns on, and it was not chosen for its looks.

| *PROVED, given the three requirements above.*

Now let the rule run, and ask what the system settles into. A self-describing system is stable when repeating the step changes its overall size but leaves the ratio between its two channels alone. It keeps its shape while it grows. Let phi stand for that ratio, resolved to held. Passing phi through the one rule returns a single equation:

$$\phi^2 = \phi + 1$$

a number whose square is merely itself plus one. There is exactly one positive solution,

$$\phi = (1 + \sqrt{5}) / 2 = 1.618...$$

the golden ratio, arriving unannounced and without having been invited. It is not there for decoration. It is the only proportion at which a two-channel system can hold its shape and go on describing itself indefinitely without gradually sliding off itself.

| *PROVED.*

The number the rest of the book actually uses comes from one further step. A single act of self-description places its mixed relation in two directions, outward to held and held to outward, so the retained side is measured with a factor of two. Define the retained amount as $r = 1 / (2\phi) = 0.309017$, and the outward amount as whatever is left, $u = 1 - r = 0.690983$, so that $u + r = 1$. One whole, divided into what a system puts out and what it keeps. No adjustable knob survives this. Once the two-sided rule is granted, u and r are pinned, with nothing left to tune.

| *PROVED, given the normalization rule.*

MOVEMENT FIVE

The three weights

Two channels, once allowed to meet, can meet in only so many ways, and the count is smaller than one might expect. Call the outward channel U and the held channel C . Taken two at a time they offer four combinations on paper: UU , UC , CU , CC . But UC and CU are the same two channels in the opposite order, and a system that does not care which way round its relations are read counts them once. Four collapses to three:

$$(u + r)^2 = u^2 + 2ur + r^2$$

Three weights, and they exhaust the whole, since they sum to one with nothing left over:

$$L = u^2 = 0.4775 \text{ (direct)}$$

$$B = 2ur = 0.4271 \text{ (related)}$$

$$M = r^2 = 0.0955 \text{ (held)}$$

This is worth sitting with. Nobody went shopping for three ingredients. Three is simply how many ways two channels can be squared together once the reading direction stops mattering.

| *PROVED.*

Then the identifications, which are a different kind of claim and are marked as one. The direct weight, content that carries with nothing held behind it, reads as light. The mixed weight, meaning that exists only in the relation between what is out and what is kept, reads as boundary. The held weight, content folded away as history, reads as matter. Light, boundary, matter: not three substances that had to be assumed, but the three ways a self-describing whole can be squared.

| *PROPOSED: the physical reading is offered for calculation and falsification, not delivered by the algebra.*

MOVEMENT SIX

The ladder

Everything so far grew from a single act of self-description. A system, though, rarely describes itself only once. It does so again, and again, and the standing question is which structures survive being described over and over.

Two operations are enough to generate them all. A system can accumulate, adding one more held distinction to what it already carries. And it can invert, folding back to make contact with itself, which is precisely what self-description is. Accumulation and inversion, taken in turn, are the whole grammar. This book calls that grammar the Distinction Algebra, and it does exactly one interesting thing: it yields a ladder of stable proportions rather than a continuum. Only certain ratios survive repeated self-description. They are the metallic means, of which the golden ratio is the first and the steadiest.

| *What earlier work named Primordial Loop Algebra is this same grammar. The old name has been retired; the two operations are unchanged.*

Of the rungs, one is an attractor. Nudge a system near the golden rung and it falls back onto it, the way a marble settles into the bottom of a bowl. That is the rung a self-maintaining system actually sits on, for the unmysterious reason that it is the one that holds. The rung just above is different in character. It is a boundary rather than a home, a proportion a system can lean against but never rest in, the rim of the bowl rather than its floor. This book calls it the silver boundary.

A careful word about that boundary, since it is the place earlier accounts tended to lose their footing. The silver boundary is the adjacent limit of the same construction, sitting just past the reach of a golden system. Nothing recordable from the inside crosses it. What lies beyond is therefore not described here, not because it is forbidden or enchanted, but because honesty about a boundary means declining to narrate the far side of it. The framework marks it as a boundary and says no more. It is emphatically not claimed to be a second universe. It is the edge of what one act of self-description can hold, which is a smaller and far more defensible thing to say.

| *POSTULATE for the boundary's role; the silence about its far side is deliberate.*

Where a golden system and the silver boundary lean against one another, they cannot stay perfectly in step. A little of the shared edge is always mismatched, always slipping. That persistent unmatched slip along the boundary is what this book names boundary shear. An earlier draft called it rubbing, a word that has since been shown the door.

MOVEMENT SEVEN

The carrier

Self-description quietly imposes one further condition, and it is strict enough to name the stage the whole drama plays out on.

A record that describes its own system has to be readable as its own dual. What the record says about the system, and what the system says through the record, have to be the same object, or the description is quietly describing something other than itself. That symmetry, self-duality, is not a soft aesthetic preference. It is a hard filter, and remarkably few structures pass it. Among the lattices that can carry the

golden proportion, the demand for an even, self-dual, minimal carrier lands on exactly one: the E8 root system, with its 240 roots. Its near relatives all fail the test, each for a reason that can be named.

| *CONDITIONAL: E8 is forced once self-duality, evenness, and minimality are required of the carrier.*

E8 is usually met as a piece of exceptional mathematics that turned up, uninvited again, in a few corners of physics. Here it is not imported from anywhere. It is what a self-describing system has to be written on. And it carries the golden proportion in its bones: project the 240 roots down to the plane, and they fall into rings that pair off in the golden ratio, agreeing to about a part in a million. The carrier and the proportion are not two pieces of luck. They are one object seen twice.

The constants the framework measures with are then read straight off this geometry, as covolumes, the sizes of the lattice's cells: $\sqrt{3}$ for the local carrier, $\sqrt{5}$ for the golden pairing, $\sqrt{8}$ for the silver boundary, 1 for E8 itself. They are not tuned to anything. They are sizes of cells in a lattice that was already fixed by the single demand to be self-dual.

MOVEMENT EIGHT

The record of the whole

From the 240-root shell down to a single number is the last descent. Summed the way the framework's own bookkeeping requires, the shell closes to one constant term, a standing readout of the vacuum, of order

$$\lambda = (4\pi / \sqrt{3}) \cdot r^{240} \approx 2.86 \times 10^{-122}$$

in natural units. The exponent is not fitted to anything. It is the 240 of the shell, doing arithmetic.

| *PROPOSED: a computed value with a stated construction, open to falsification.*

Underneath the number is the thing the whole story has been walking toward, which is a direction. Look once more at the one rule. The next state always carries the present resolution plus everything held, and what is held only ever grows, because to keep describing itself a system must keep more of itself in reserve. Held action is not lost. Held action is structure. What accumulates as a system goes on describing itself is exactly what gets called, from outside and after the fact, complexity.

This is the causal arrow, and it is worth being exact about where it comes from. It does not come from anything running down, or wearing out, or being paid for. It comes from the plain fact that self-description adds: each pass keeps what the last one resolved. A system that describes itself cannot help but accumulate, and accumulation has a direction built into it, since what is already held cannot be unheld. Time, told this way, is not a stage the system stands on. It is the shape of the system's own record, growing. The arrow points the way it does for the simple reason that reserves only fill.

| *CONDITIONAL: the direction follows from the additivity of the update. Its identification with physical time is PROPOSED.*

MOVEMENT NINE**What it claims, and what it leaves open**

A book that derives this much owes a plain account of what has actually been earned. So the claims are kept in grades, and the grades are never quietly folded into one another. A mathematical result that follows from stated assumptions is one kind of thing. A physical reading laid on top of it is a different kind of thing, and no amount of enthusiasm converts the second into the first. The golden proportion and the three weights are proved. That E8 is the carrier is conditional on the demands placed on the carrier. That the weights are light, boundary and matter is proposed, and lives or dies by measurement. Keeping these apart is not modesty for its own sake. It is the difference between a framework and a mood.

An example, kept deliberately small. Water. Two bonds of hydrogen held through one of oxygen, closing at an angle near a hundred and five degrees, leaving a record in every spectrum it touches. The framework does not predict water into existence. It describes water in its own terms, distinguish and hold and close and record, and shows that the same seven doings that opened the book run just as cleanly through a molecule as through a cosmos. The example is not evidence. It is a demonstration that the language does not crack when pointed at something ordinary.

And then the book stops, on purpose, a step short of the reader. What a self-describing universe of this kind finally means, whether the golden proportion in a chain of atoms and the golden proportion in the shell are one fact wearing two coats, what sits past the silver boundary: these are left open, not because the framework has nothing to say, but because a language earns trust by describing precisely and declining to oversell. The sentence set out to describe itself. It has. Where that leads is left, as it ought to be, to whoever reads it next.